





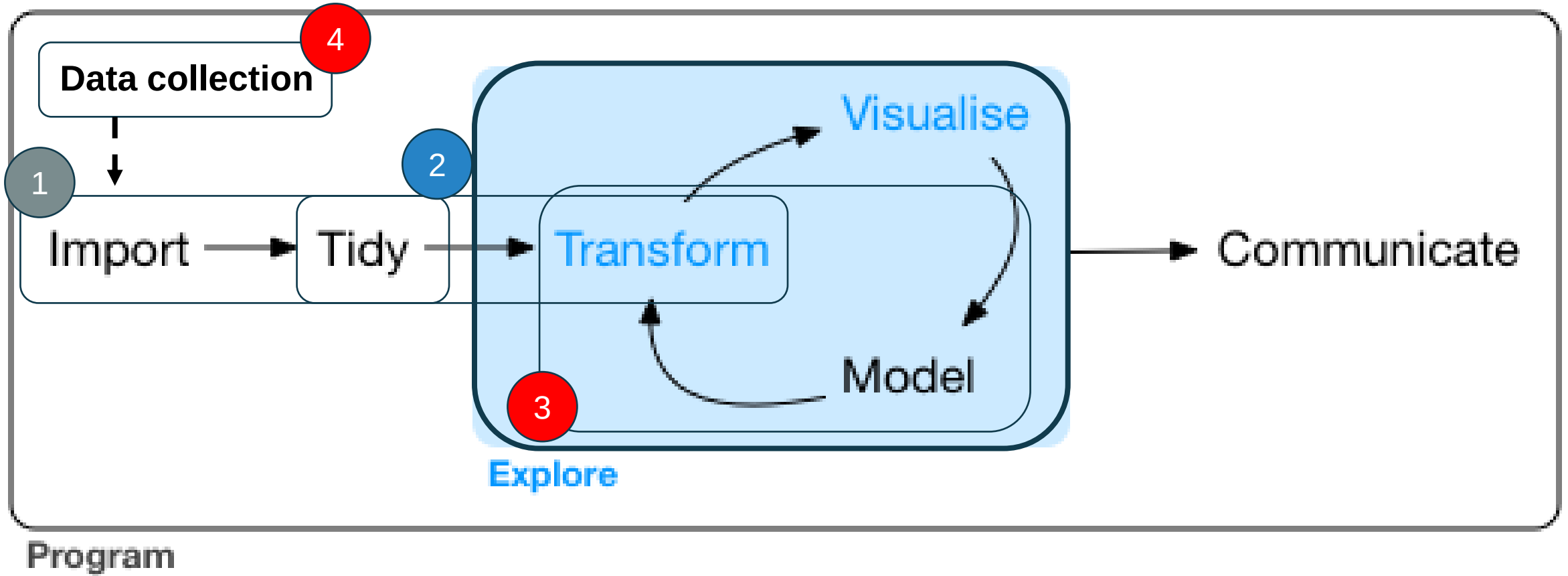
ENVRSUST 9014

Environmental Data Analysis and Visualization

Dr Paul Mensink



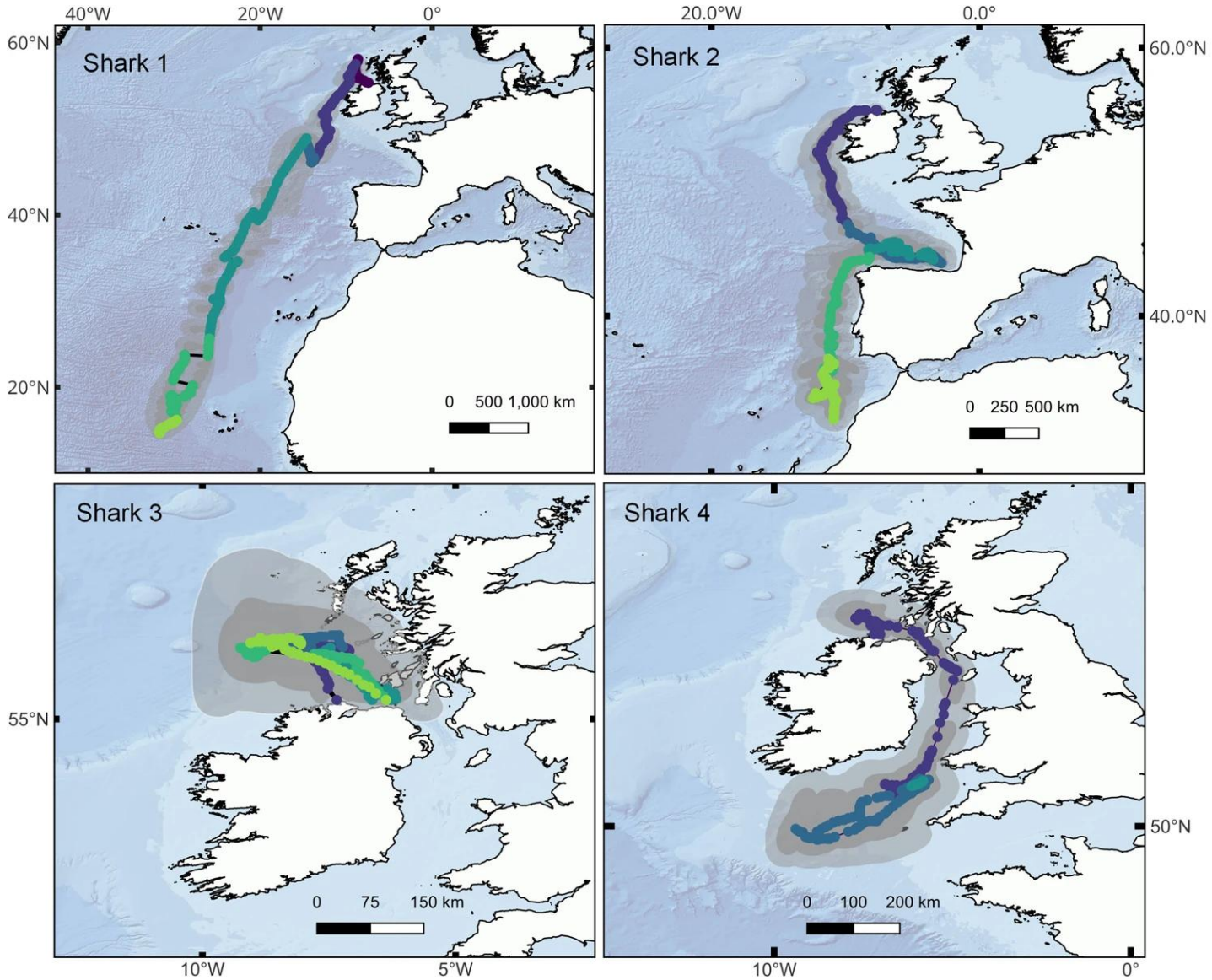




R for Data Science
https://r4ds.had.co.nz/index.html

Take-up and review Topic 1

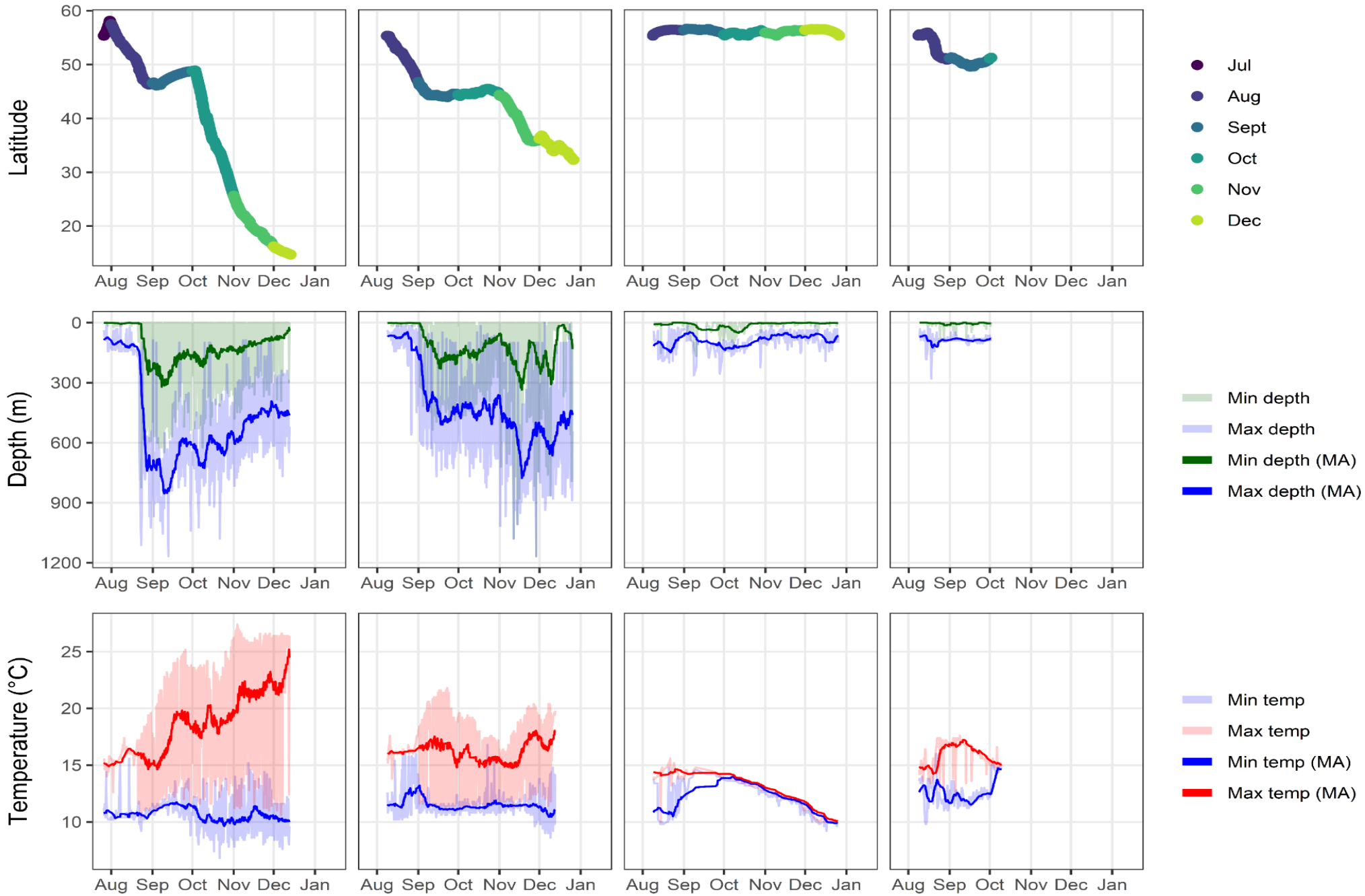
Sourcing open data

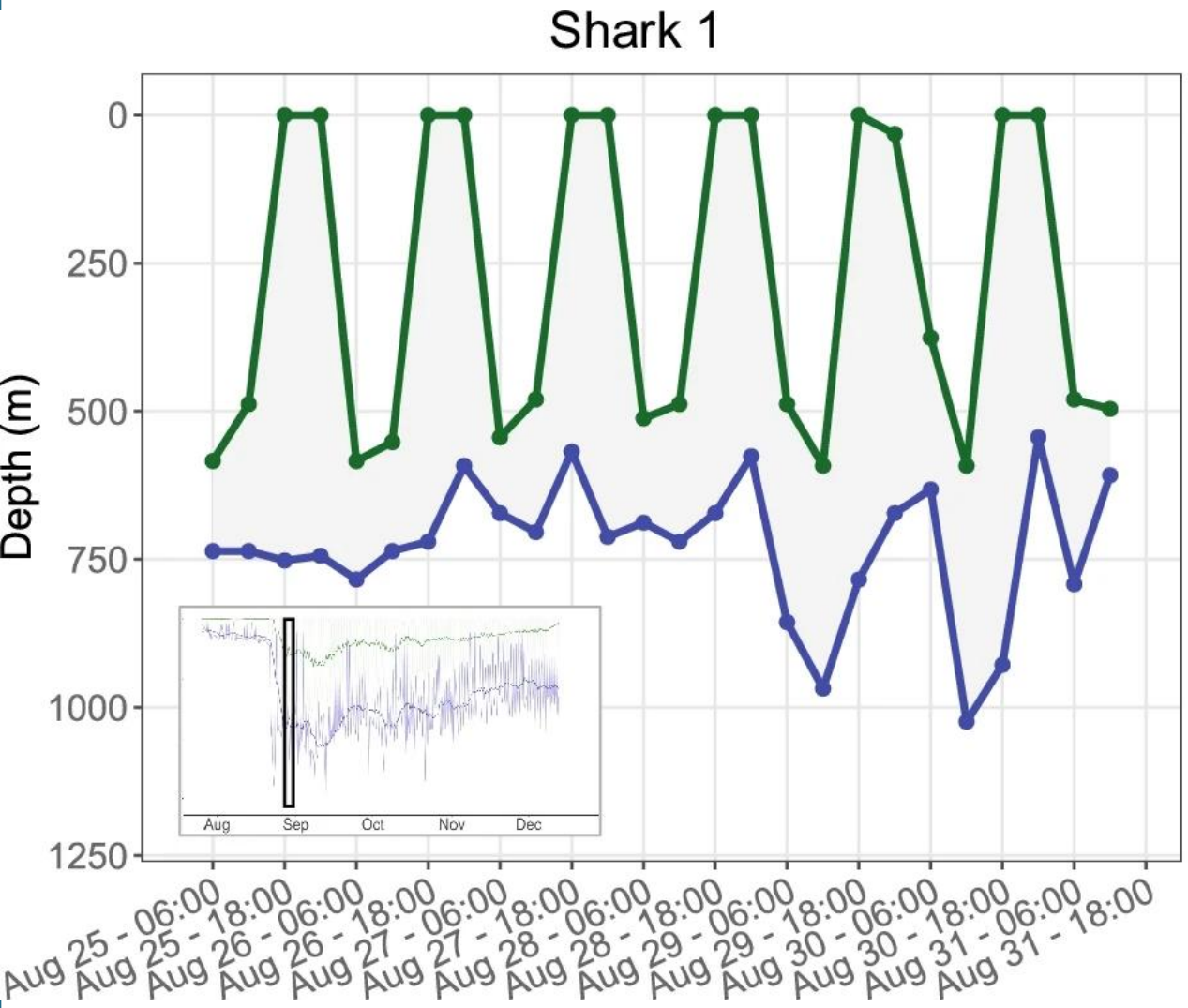


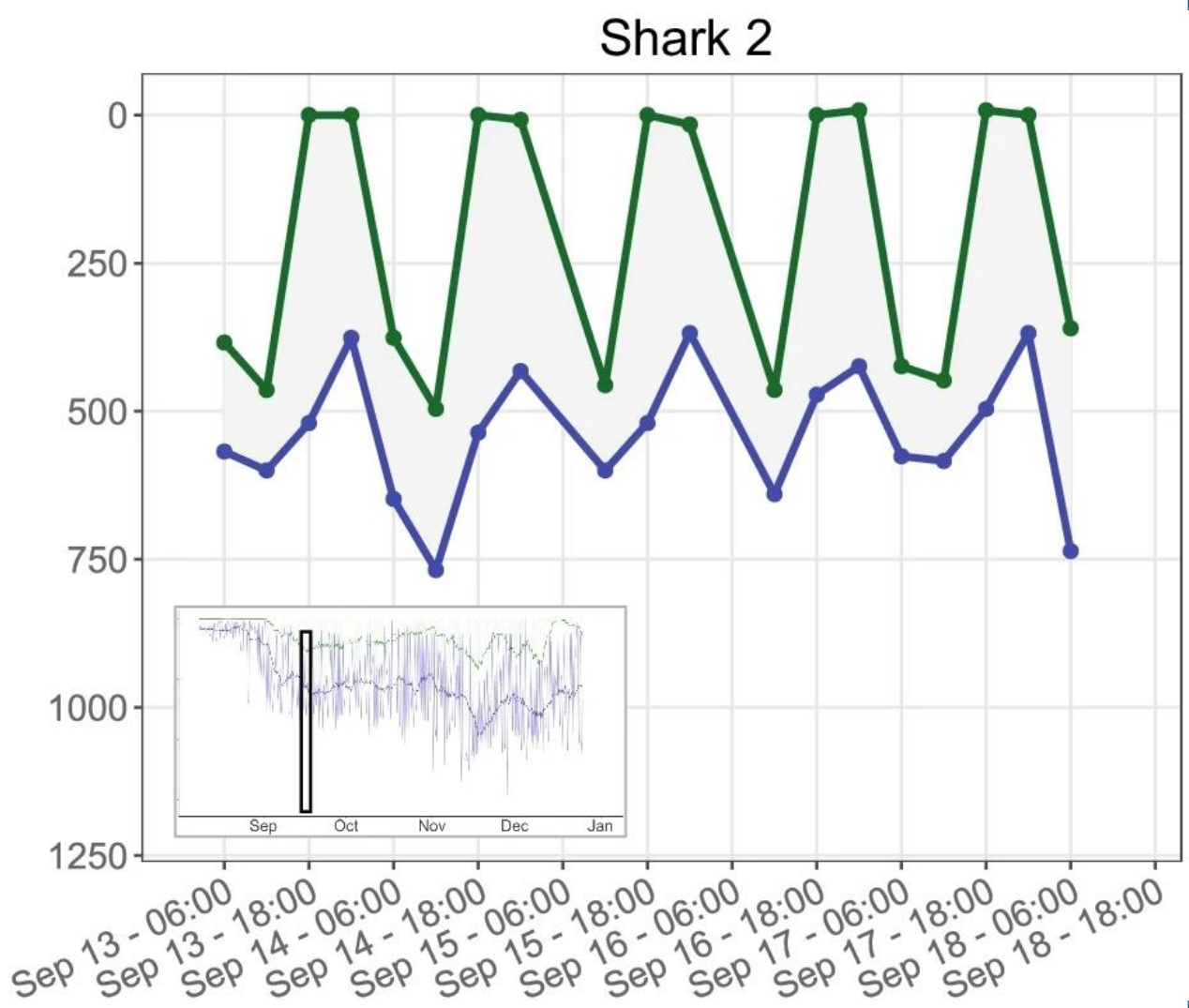
Environ Biol Fish
https://doi.org/10.1007/s10641-021-01202-8

Cool runnings: behavioural plasticity and the realised thermal niche of basking sharks

E. M. Johnston · J. D. R. Houghton · P. A. Mayo · G. K. F. Hatten · A. P. Klimley · P. J. Mensink

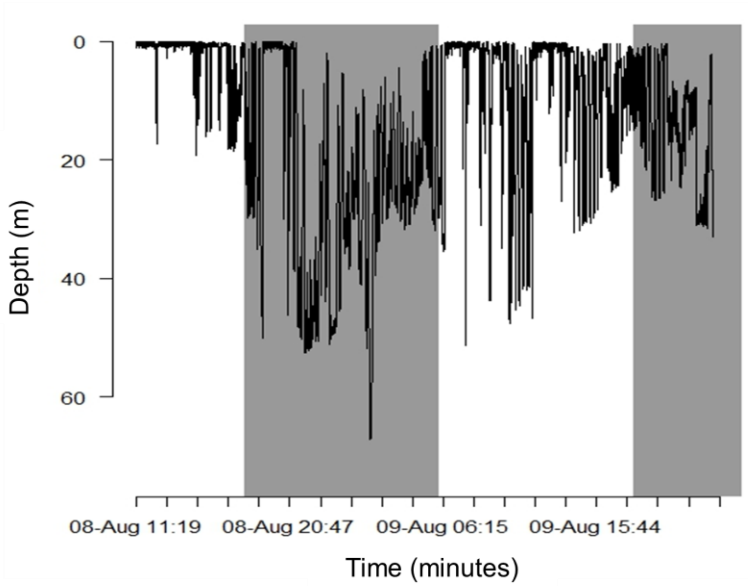

Latitude
60
50
40
30
20
Aug Sep Oct Nov Dec Jan
Jul
Aug
Sept
Oct
Nov
Dec
Depth (m)
0
300
600
900
1200
Min depth
Max depth
Min depth (MA)
Max depth (MA)
Temperature (°C)
25
20
15
10
Min temp
Max temp
Min temp (MA)
Max temp (MA)


Shark 1
Depth (m)
0
250
500
750
1000
1250
Aug
Sep
Oct
Nov
Dec
Aug 25 - 06:00
Aug 25 - 18:00
Aug 26 - 06:00
Aug 26 - 18:00
Aug 27 - 06:00
Aug 27 - 18:00
Aug 28 - 06:00
Aug 28 - 18:00
Aug 29 - 06:00
Aug 29 - 18:00
Aug 30 - 06:00
Aug 30 - 18:00
Aug 31 - 06:00
Aug 31 - 18:00


Shark 2
0
250
500
750
1000
1250
Sep
Oct
Nov
Dec
Jan
Sep 13 - 06:00
Sep 13 - 18:00
Sep 14 - 06:00
Sep 14 - 18:00
Sep 15 - 06:00
Sep 15 - 18:00
Sep 16 - 06:00
Sep 16 - 18:00
Sep 17 - 06:00
Sep 17 - 18:00
Sep 18 - 06:00
Sep 18 - 18:00

1

Collect data on depth profiles using time-depth recorders (n=13 sharks)



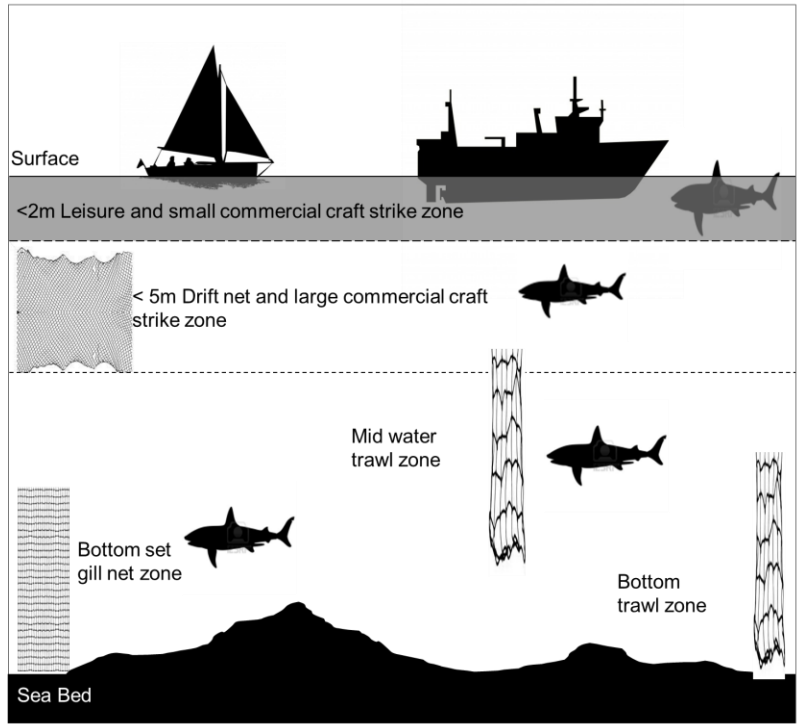
Assign states based on relevant threats

S_1

Vulnerable to ship strikes

S_2

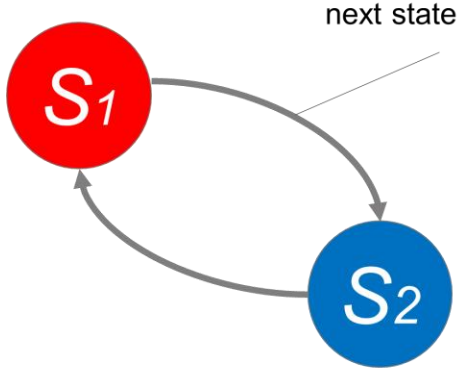
Safe from ship strikes



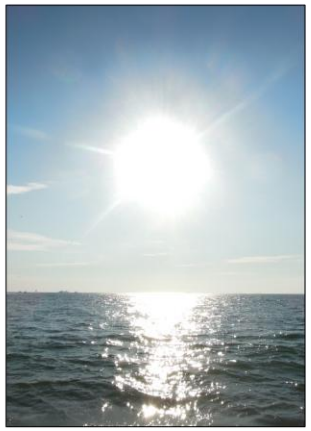
2

Conceptualize a 2-state semi-Markov model

Transition intensities (time spent in current state and probability of next state occurring)



Obtain irradiance data for the time-depth recorder deployments

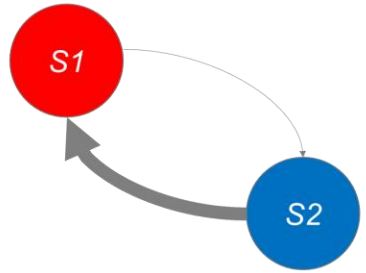




Weather data can be easily obtained and downloaded retrospectively for previous time-depth recorder deployments from Met Éireann and other data sources

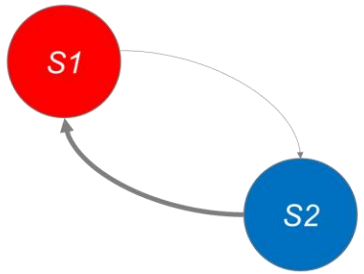
Condition transition intensities on irradiance

More likely to be at the surface under high irradiance



Hazard ratio from state 2 to state 1: **12.01** (95% CI = 10.39, 13.88)

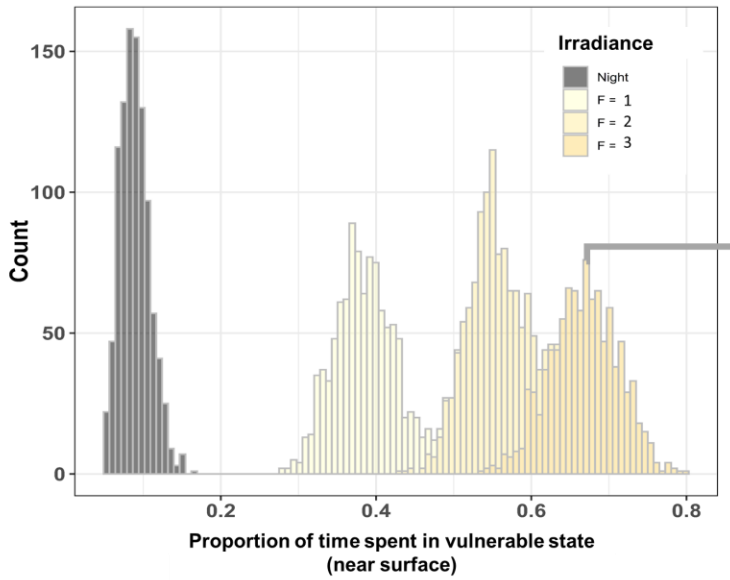
Less likely to be at the surface when irradiance is low



Hazard ratio from state 2 to state 1: **4.31** (95% CI = 3.76, 4.93)

3

Simulate the proportion of time spent in a vulnerable state



(60 minute period, N_{sims} = 1000)

Provide adaptive and dynamic real-time management for stakeholders

Likelihood of boat strike low

Likelihood of boat strike high, reduce boat speed and avoid known hotspots

Predictions can update with real-time changes in weather

Deploy information through phone app and website

Inform recreational boat users about high risk

Instruct commercial vessels to reduce speed

Suggest set times for fishing nets

Recommend preferred times for shoreline ecotourism

Capture, Mark & Recapture (CMR)

- Animal captures & ID tag attached
- Recaptured at later date & ID noted
- Two known points of information
- CMR can identify:
  - site fidelity (to some extent)
  - long distance movement
  - mortality rates & population counts



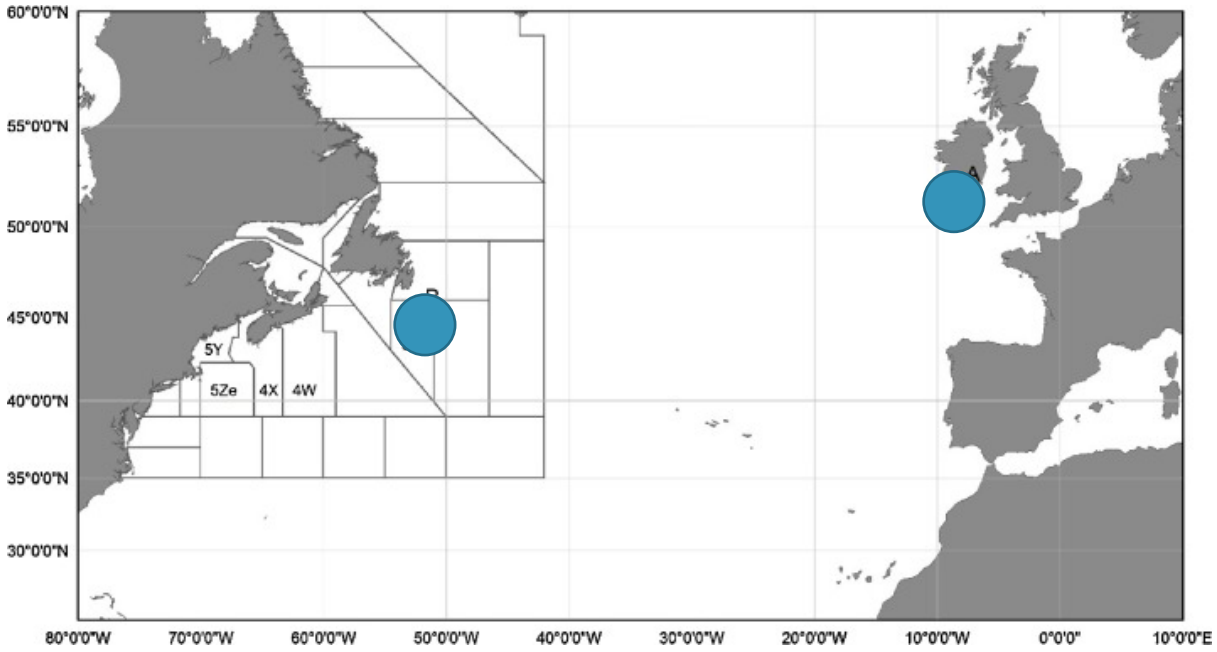


Capture, Mark & Recapture (CMR)

Marine Sportfish tagging system

- “Over 40,000 individual fish have been caught, measured, tagged and released over 50 years with over 3,000 recaptures.”





ICES Journal of Marine Science

ICES Journal of Marine Science (2019), 76(6), 1581–1590. doi:10.1093/icesjms/fsz046

Original Article

Population structure and spatial distribution of porbeagles (*Lamna nasus*) in Irish waters

Luke W. J. Cameron [1,2,3], William K. Roche[2,*], Jonathan D. R. Houghton[1,4], and Paul J. Mensink[1,5]

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